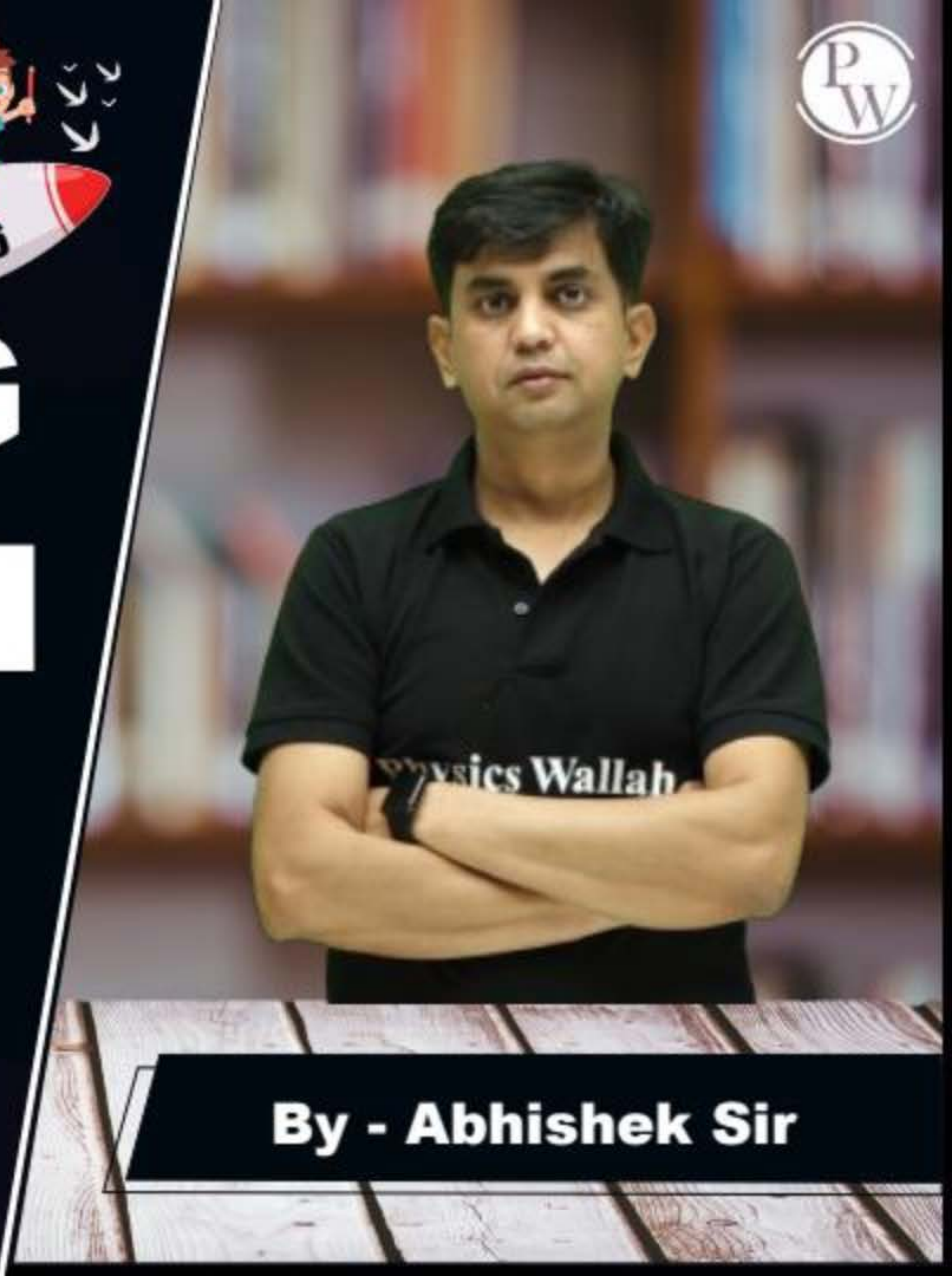
CS & IT ENGINEERING

Computer Network

IPv4 Header

Lecture No. - 02



By - Abhishek Sir

Recap of Previous Lecture

Topic IPv4 Header Format

Topic IPv4 Header and Packet Size

Topic MTU

Topics to be Covered

Topic Identification Number

Topic Fragmentation at Source Host

ABOUT ME

Hello, I'm Abhishek

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Topic : IPv4 Packet Header

words of 4 bytes

0			16				31
VER	[HLEN]	Type of Services	[Total Length] bytes				
Identification Number			0	DF	MF	[Fragmentation Offset]	words of 8 bytes
Time-to-Live		Protocol Type	Header Checksum				
Source IPv4 Address (32 - bits)							
Destination IPv4 Address (32 - bits)							
Optional Header (Options)							
Payload							

Topic : Identification Number

→ 16 bits long

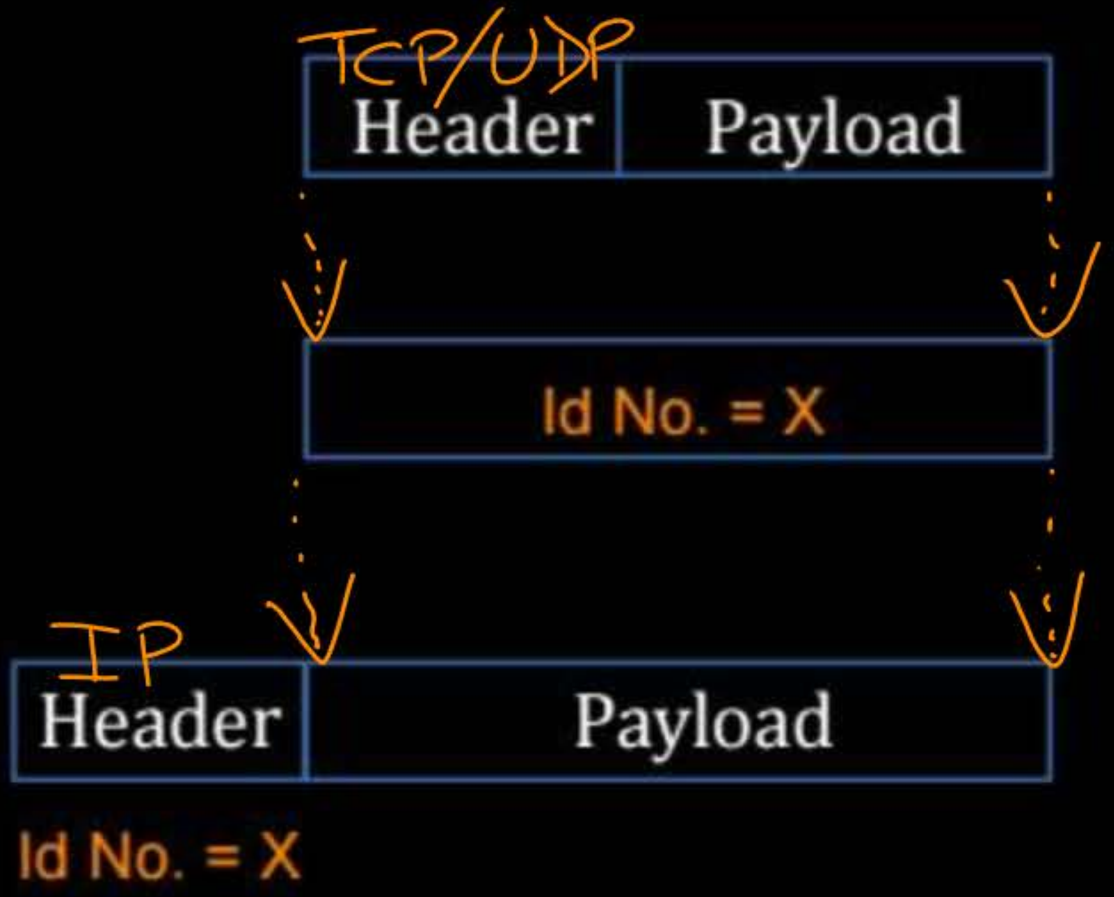
→ Assigned by source host only
[Assigned unique identification number to each transport layer Segment]

→ IP fragments of same segment, must have same identification number
[does not change during routing]

Topic : Identification Number

Transport Layer PDU (Segment)

SDU for Network Layer



Topic : Identification Number

TCP/UDP segment

Transport Layer PDU (Segment) | Header | Payload

SDU for Network Layer | Id No. = Y

IP Header	Payload_1

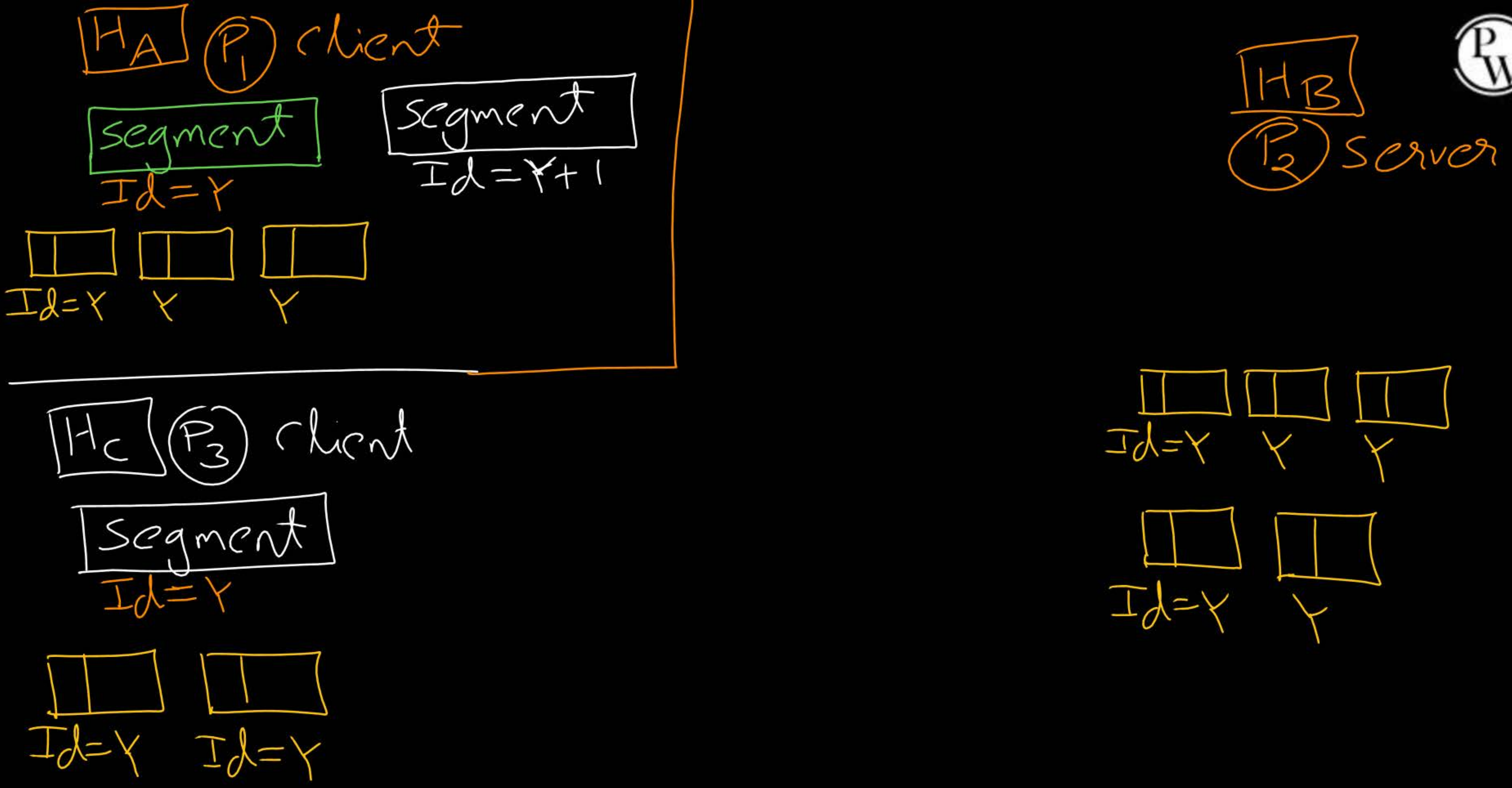
Id No. = Y

IP Header	Payload_2

Id No. = Y

IP Header	Payload_3

Id No. = Y



H_A
P_1 client
segment
Id = Y
segment
Id = Y+1
Id = Y
Y
Y
H_B
P_2 server
Id = Y
Y
Y
Id = Y
Y
H_C
P_3 client
segment
Id = Y
Id = Y
Id = Y

Topic : Identification Number

→ Receiver host do clustering of fragments, before assembly

→ based on Source IP Address and Identification number

Identification Number (16 bits) : Locally unique

Source IP Address & Identification Number (48 bits) : Globally unique

Example 6 :- [NAT]

#Q. Consider each host in an IPv4 network uses [up to 1024 unique identifiers per second. After what period (in seconds) will the identifiers generated by a host wrap around ?

Identification No.

$$= \underbrace{0, 1, 2, 3, \ldots\ldots (2^{16}-1)}_{2^{16} \text{ numbers}}, 0, 1, 2, \ldots\ldots$$

$$\boxed{Ans = 64}$$

Identification Number : 16 bits

Total number of unique identifiers = $\boxed{2^{16}}$
[In each host, before wrap-around]

1024 identifiers → 1 Sec

2^{16} identifiers → $\frac{2^{16}}{2^{10}}$ sec $= 2^6$ sec $= 64$ sec

[NAT] (H.W.) IIT-KGP [GATE-2014] [2 Mark]

#Q. Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

Example 7 :- [V. Imp.] **[NAT]**

#Q. Consider UDP segment of size 4000 bytes is passed to IPv4 protocol for delivery. MTU for source network is 1500 bytes and IPv4 header size is 20 bytes then calculate total number of fragments required to deliver the UDP segment ?

[IP Packet Size ≤ MTU]

IP Packet Size = [Header Size + Payload Size]

$$\boxed{\text{Ans} = 3}$$

MTU = 1500 bytes

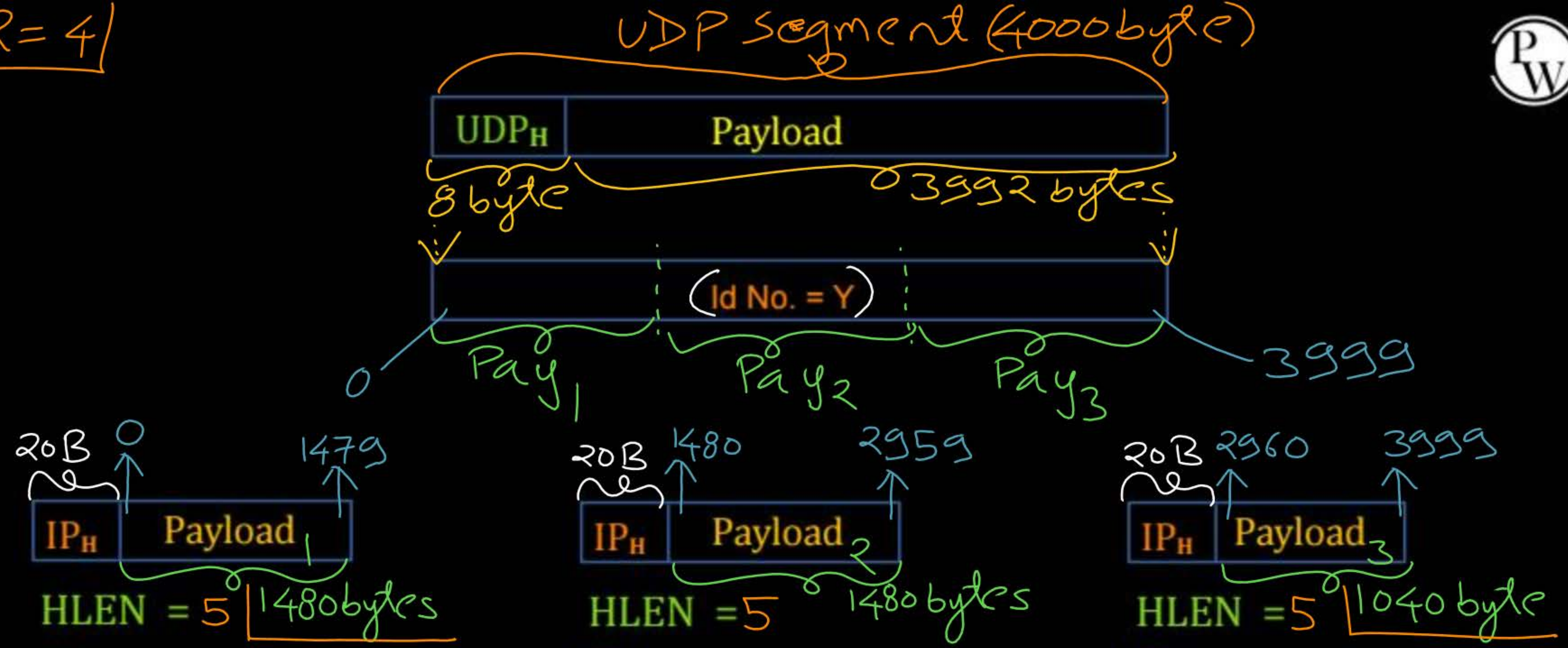
IPv4 Header Size = 20 bytes

$$\boxed{\text{Maximum Payload Size} = [\text{MTU} - \text{Header Size}]\ \text{bytes}}$$

$= [1500\ \text{byte} - 20\ \text{byte}]$

$= 1480\ \text{bytes}$

$\boxed{VER = 4}$



Fragment 1	Fragment 2	Fragment 3
HLEN = 5	HLEN = 5	HLEN = 5
TL = 1500	TL = 1500	TL = 1060
Id No. = Y	Id No. = Y	Id No. = Y
Offset = $\frac{0}{8} = 0$	Offset = $\frac{1480}{8} = 185$	Offset = $\frac{2960}{8} = 370$
MF bit = 1	MF bit = 1	MF bit = 0

UDP Segment Size = (4000 bytes)

$$\text{Total Number of IP fragments } [N] = \left\lceil \frac{\text{UDP Segment Size}}{\text{Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{4000 \text{ bytes}}{1480 \text{ bytes}} \right\rceil$$

$$= \lceil 2.70 \rceil$$

$$N = 3$$

Total length of the last IP fragment =

= Header Size + [UDP Segment Size – (N–1) * Payload Size] bytes

$= 20 \text{ byte} + [4000 \text{ bytes} - (3-1) * 1480 \text{ bytes}]$

$= 20 + [4000 - 2960] \text{ bytes}$

$= (20 + 1040) \text{ bytes}$

$= 1060 \text{ byte}$

$$\text{Offset value of the last IP fragment} = \left[\frac{(N-1) * \text{Payload Size}}{8}\right]$$

$$= \left[\frac{(3-1) * 1480 \text{ bytes}}{8}\right]$$

$$= \left[\frac{2960 \text{ byte no.}}{8}\right]$$

$$= 370 \text{ word no.}$$

Topic : Fragmentation Offset

→ Fragmentation offset is 13 bits long

→ Used to identify the sequence of fragments

→ Contains payload's starting word number [Words of 8 bytes]

→ Word number according to "Service Data Unit" (SDU)

→ Offset value for the first fragment in the sequences always "Zero"

Byte No.

$$\left.\begin{matrix}0\\1\\2\\3\\4\\5\\6\\7\end{matrix}\right\}\text{word}_0 = \frac{0}{8}$$

$$\left.\begin{matrix}8\\9\\\vdots\\15\end{matrix}\right\}\text{word}_1 = \frac{8}{8}$$

$$\left.\begin{matrix}16\\\vdots\\23\end{matrix}\right\}\text{word}_2 = \frac{16}{8}$$

$$\left.\begin{matrix}24\\\vdots\\31\end{matrix}\right\}\text{word}_3 = \frac{24}{8}$$

Topic : 3 Flag Bits

→ First bit unused, must be zero

→ DF : Do not fragment

→ MF : More fragment

Topic : MF Flag

MF : More fragment

→ Used to identify last fragment in the sequence of fragments

→ For last fragment, MF bit should be "Zero"

→ For first and intermediate fragments (except last), MF bit should be "One".

Topic : Fragmentation at Source Host

IPv4 Datagram Size ≤ Source Network MTU

Maximum Payload Size = [MTU – Header Size] bytes

Total Number of IP fragments [N] = ⌈[UDP Segment Size / Payload Size]⌉

Total length of the last IP fragment

= Header Size + [UDP Segment Size – (N–1) * Payload Size] bytes

Offset value of the last IP fragment = [(N–1) * Payload Size / 8]

[MCQ] H.W. IIT-B [GATE-2013] [2 Mark]

#Q. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

A. Last fragment, 2400 and 2789

B. First fragment, 2400 and 2759

C. Last fragment, 2400 and 2759

D. Middle fragment, 300 and 689

[MCQ], H.W. IIT-K [GATE-2015] [2 Mark]

#Q. Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How may total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

A 6 and 925

B 6 and 7400

C 7 and 1110

D 7 and 8880

[MCQ] , H.W. IIT-R [GATE-2025] [1 Mark]

#Q. Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?

(A) 5 fragments, 1488 bytes

(B) 6 fragments, 88 bytes

(C) 6 fragments, 108 bytes

(D) 6 fragments, 116 bytes

2 mins Summary

Topic Identification Number

Topic Fragmentation at Source Host

THANK - YOU

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